

Pedda Pullannagari Sreedhar

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PROFESSIONAL SUMMARY

- Full Stack Developer focused on backend systems that stay correct under load - idempotent workflows, cached read paths, and role-scoped APIs - with React and TypeScript on the frontend.

EXPERIENCE

Full Stack Developer

Nov 2024 -- Present

Orcalex Technologies LLP

Hyderabad, India

- Engineer SmartLearners.ai / AiEducator, a multi-school EdTech platform serving 10k+ students across schools (B2B) and individual learners (B2C), on Django/DRF, FastAPI, and React.
- Built the auth layer for admin, school-admin, teacher, and student roles: header JWT for mobile clients, cookie auth for web, OAuth/OIDC sign-in, and JWT-authenticated WebSockets.
- Developed the live classroom module: homework and scheduled exams persist to the database, push to online students over WebSockets, and replay as unread notifications when offline students reconnect.
- Fixed a cache stampede there - one expiring Redis key sent ~500 reconnecting students to the database - with double-checked locking, collapsing the burst into a single query.
- Shipped the question bank, worksheet, and scheduled-assignment workflows, owning the REST APIs and the React state behind them.
- Grew AI evaluation from a single autoscore service to four - autoscore, homework, classwork, and exam correction - dispatching uploads over Celery and persisting scores, gap analysis, and class analytics as domain state.
- Cut API response times ~30% with EXPLAIN ANALYZE profiling, composite indexes, and Redis caching over optimized ORM queries.
- Wrote unit and integration tests that gate every PR, with main deploying to staging and tags promoting to production.
- Operate production on DigitalOcean Kubernetes (HTTP, WebSocket, Celery worker and beat deployments, plus FastAPI exam-correction workers) and on Linux VMs behind Nginx, with PgBouncer pooling Postgres connections.

PROJECTS

SmartLearners.ai | React, Django REST Framework, Celery, Redis, PostgreSQL

- Migrated the platform from single-school with boolean role flags to multi-school, multi-section, role-scoped access.
- Zero-downtime schema changes: nullable columns first, dual-write, batched backfill, then NOT NULL.
- Resolved every request to an authorized school set - platform admin, school admin, or multi-school manager - before any read or write.
- Made every bulk and destructive operation dry-run first, with per-row validation and structured errors before commit.
- Cached school reads under role-aware versioned keys, so one version bump invalidates a whole tenant's cache without scanning keys.
- Instrumented audit and critical-event logging so admins can trace who changed exams, results, and school data.
- Moved question and answer images from database rows into object storage with Django-generated presigned uploads.
- Integrated PhonePe payments: order creation, webhook verification, status polling, retry, refund, and activation.

Exam Correction Service | FastAPI, SQLAlchemy, Django, Celery, Redis, PostgreSQL

- Built the runtime and persistence layer of the FastAPI correction service for group (up to 50 students) and single-student runs.
- Kept all identity in Django: the service holds no auth, uploads results to object storage, and posts to a webhook that persists them for the owning teacher only.
- Designed the lifecycle around one idempotency key shared by Django, Celery, and FastAPI, with teachers seeing live phase and percent complete.
- Defined the bi-directional webhook contracts and a batch-recovery API that restores in-flight corrections after logout or refresh.
- Normalized and matched roll numbers so only verified students persist, then recalculated exam totals, averages, and class analytics, with token usage, cost, and unmatched rolls in an admin-only audit model.
- Added an admin retry action that re-queues failed batches from saved question-paper and answer-sheet snapshots.

Coordinator Admin Dashboard | React, TypeScript, Vite, Tailwind, Bun

- Sole developer of a nine-panel school-admin dashboard in React and TypeScript: school data, student and teacher management, class-section access, reassignment, archival, and record merging.
- Gated every destructive action behind a dry-run preview whose approval signature must match before the commit is allowed.
- Implemented dual JWT/cookie auth with single-flight token refresh, plus a hand-rolled CSV parser for bulk import and export.
- GitHub Actions: config validation, lint, build, then rsync deploy to the VM with an Nginx validate-and-reload step.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript

Frontend: React, TypeScript, Vite

Backend: Django, Django REST Framework, FastAPI, Django Channels

Databases & Caching: PostgreSQL, MySQL, Django ORM, SQLAlchemy, Redis, Celery

Infrastructure: L7 proxies, AWS (EC2, S3, Lambda, RDS, DynamoDB), DigitalOcean (Spaces), Ubuntu

Containers & Orchestration: Docker, Kubernetes (Deployments, Services, Ingress, Jobs)

APIs & Tools: REST APIs, WebSockets, JWT & RBAC, OAuth/OIDC, PSP and messaging integrations, Git, GitHub Actions, CI/CD

EDUCATION

JNTUA College of Engineering

Aug 2020 -- May 2024

Bachelor of Technology in Computer Science and Engineering; GPA: 7.2

Kalikiri, AP